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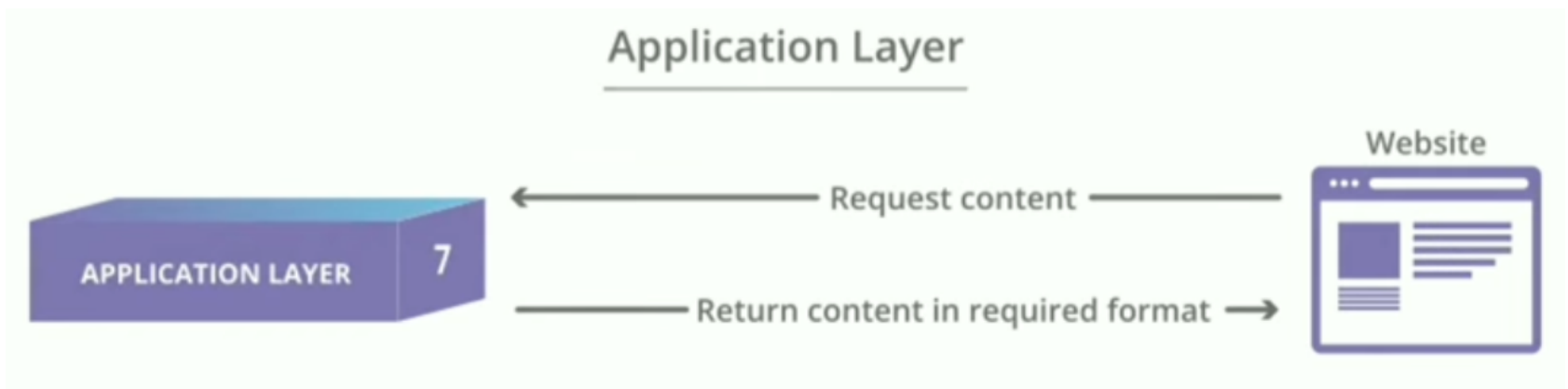
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APPLICATION LAYER

Application Layer

- The application layer is responsible for providing services to the user.
- The client server application programs can be divided into two categories
- Those that can be directly used by the user, such as email and those that support other application programs.



What is a protocol?

- In diplomatic circles, a protocol is the set of rules governing a conversation between people
- We have seen that the client and server carry on a machine-to-machine conversation
- A network protocol is the set of rules governing a conversation between a client and a server
- There are many protocols, HTTP is just one

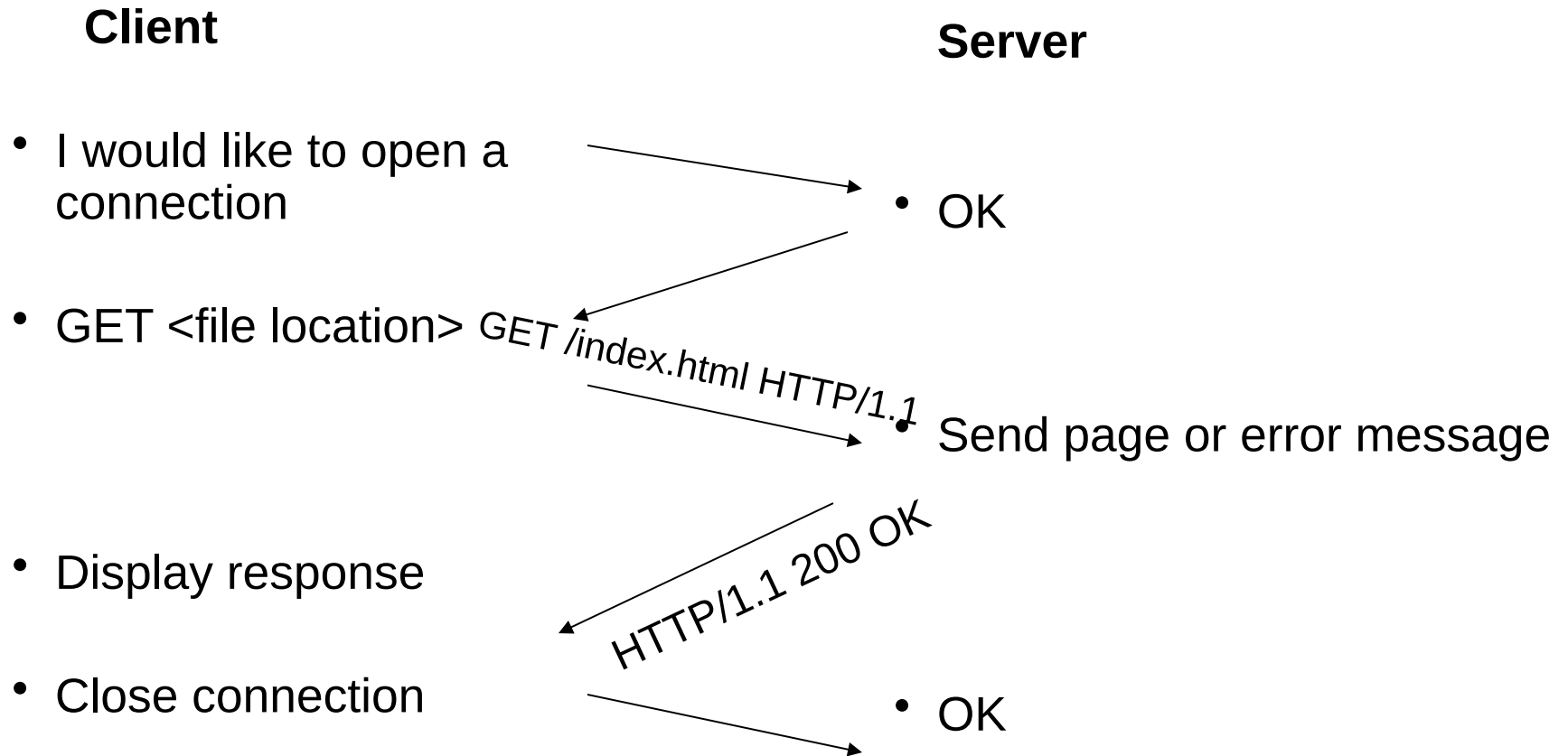
HTTP vs HTML

- HTML: hypertext **markup language**
 - Definitions of tags that are added to Web documents to control their appearance
- HTTP: hypertext transfer **protocol**
 - The rules governing the conversation between a Web client and a Web server

Both were invented at the same time by the same person

:Tim Berners-Lee

An HTTP conversation

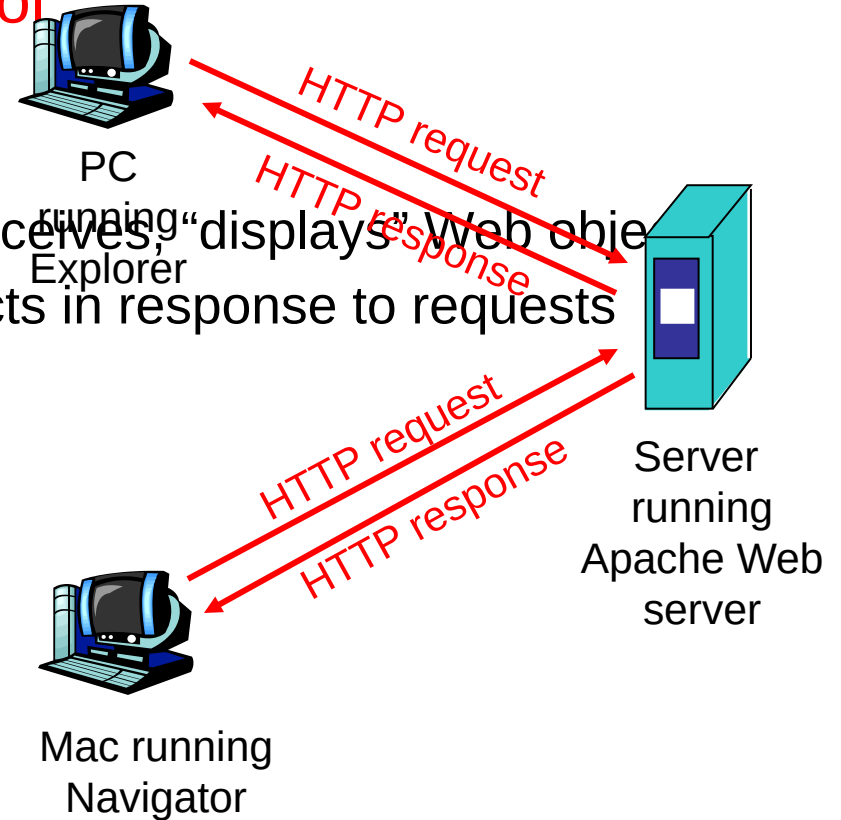


HTTP is the set of rules governing the format and content of the conversation between a Web client and server

HTTP overview

HTTP: hypertext transfer protocol

- Web's application layer protocol
- client/server model
 - *client*: browser that requests, receives, "displays" Web objects
 - *server*: Web server sends objects in response to requests
- HTTP 1.0
- HTTP 1.1



Web and HTTP

First some jargon

- **Web page** consists of **objects**
- Object can be HTML file, JPEG image, Java applet, audio file,...
- Web page consists of **base HTML-file** which includes several referenced objects
- Each object is addressable by a **URL**
- Example URL:

`www.someschool.edu/someDept/pic.gif`

host name

path name

HTTP overview (continued)

Uses TCP:

- client initiates TCP connection (creates socket) to server, port 80
- server accepts TCP connection from client
- HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP is “stateless”

- server maintains no information about past client requests

aside

Protocols that maintain “state” are complex!

- ☐ past history (state) must be maintained
- ☐ if server/client crashes, their views of “state” may be inconsistent, must be reconciled

HTTP connections

Nonpersistent HTTP

- Multiple objects are sent over a single connection
- between client and server
- HTTP/1.0 uses nonpersistent HTTP
- HTTP/1.1 uses persistent connections in default mode

HTTP request message

- two types of HTTP messages: *request, response*
- **HTTP request message:**
 - ASCII (human-readable format)

The diagram illustrates the structure of an HTTP request message. It consists of a request line followed by header lines, and a final carriage return and line feed. Annotations with arrows point to these components: 'request line (GET, POST, HEAD commands)' points to the first line; 'header lines' points to the subsequent lines; and 'Carriage return, line feed indicates end of message' points to the final line.

```
GET /somedir/page.html
HTTP/1.1
Host: www.someschool.edu
User-agent: Mozilla/4.0
Connection: close
Accept-language: fr
```

(extra carriage return, line feed)

Uploading form input

Post method:

- Web page often includes form input
- Input is uploaded to server in entity body

URL method:

- Uses GET method
- Input is uploaded in URL field of request line:

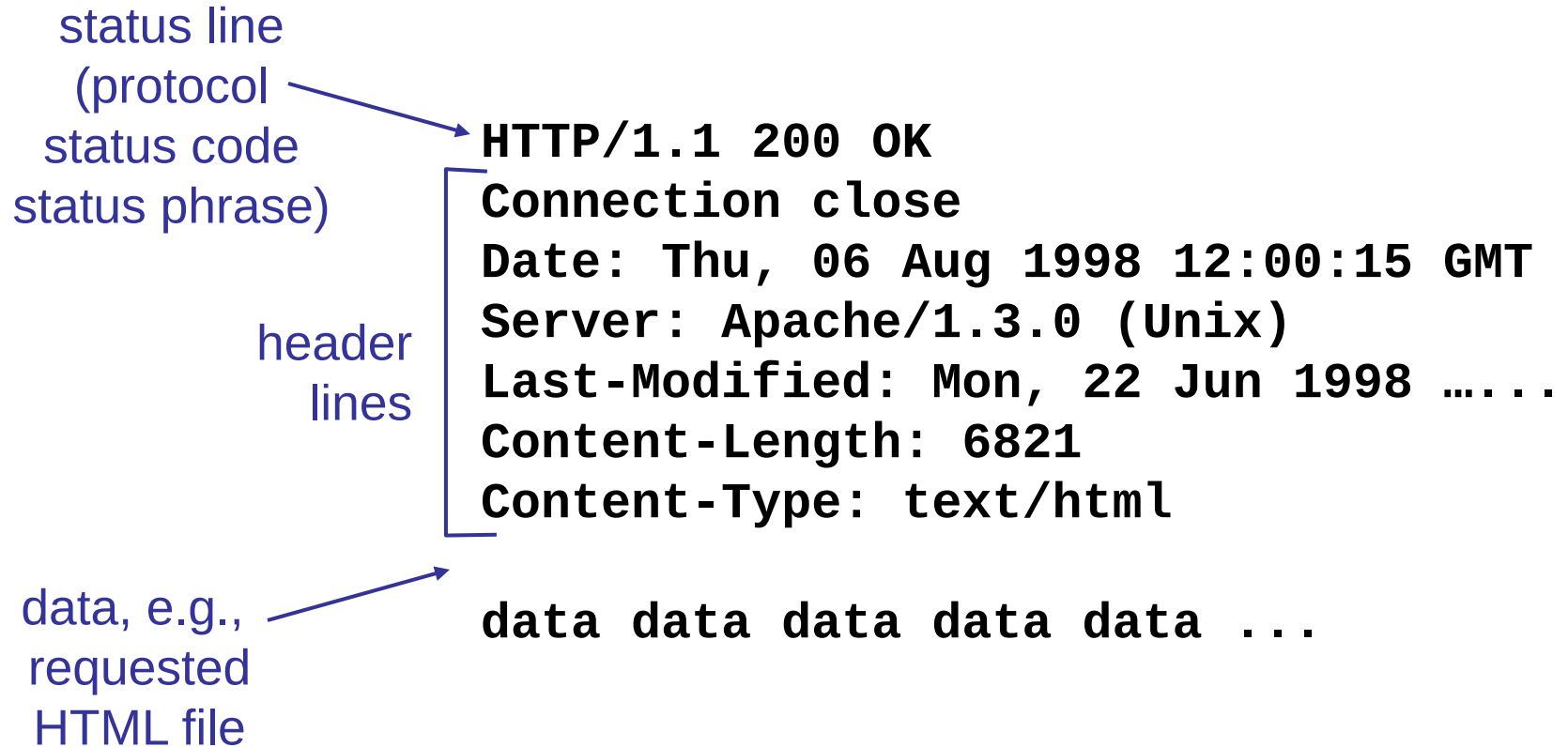
`www.somesite.com/animalsearch?
monkeys&banana`

Method types

HTTP/1.0

- GET, POST, HEAD
- POST
- HEAD
- DELETE
 - deletes file specified in the URL field

HTTP response message



HTTP response status codes

In first line in server->client response message.

A few sample codes:

200 OK

- request succeeded, requested object later in this message

301 Moved Permanently

- requested object moved, new location specified later in this message (Location:)

400 Bad Request

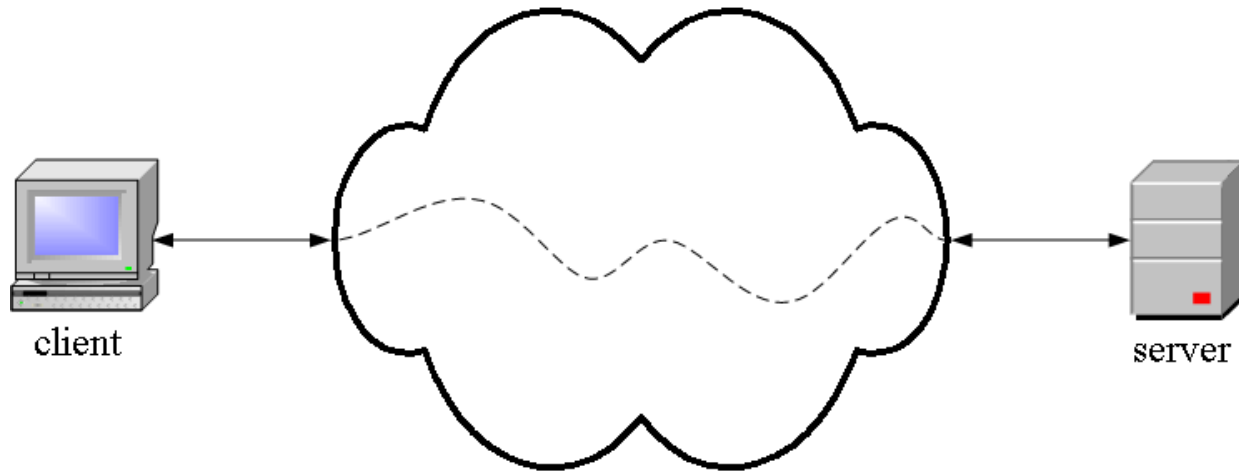
- request message not understood by server

404 Not Found

- requested document not found on this server

505 HTTP Version Not Supported

HTTP is an **application layer** protocol



- The Web client and the Web server are application programs
- Application layer programs do useful work like retrieving Web pages, sending and receiving email or transferring files
- Lower layers take care of the communication details
- The client and server send messages and data without knowing anything about the communication network

Many application layer protocols are used on the Internet

Protocol	Application
HTTP: Hypertext Transfer	Retrieve and view Web pages
FTP: File Transfer	Copy files from client to server or from server to client
SMTP: Simple Mail Transport	Send email
POP: Post Office	Read email

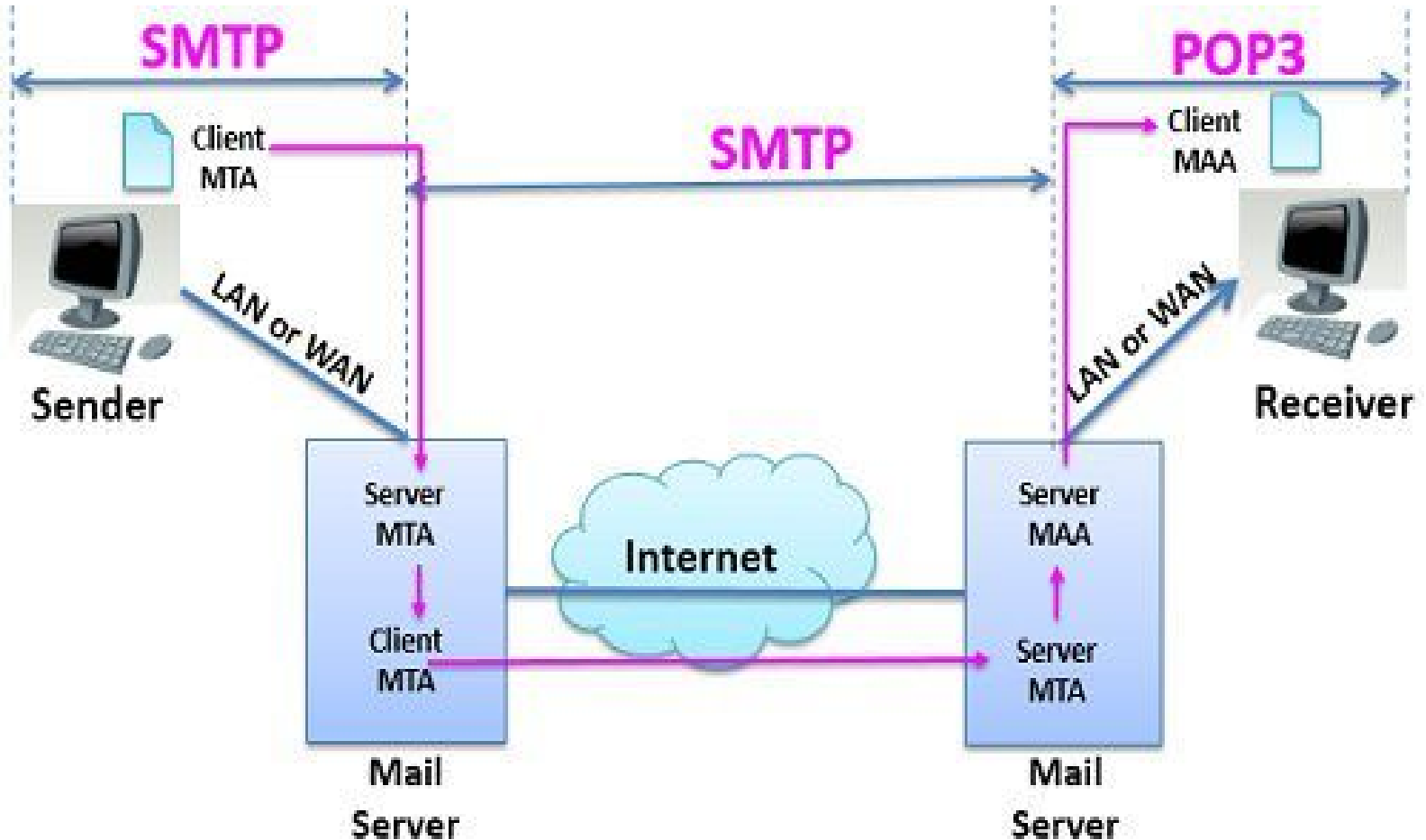
SMTP - POP3 - IMAP

We have one *message transfer agent* i.e. **SMTP** (Simple Mail Transfer Agent), and we have two *message access agents* **POP** (Post Office Protocol) and **IMAP** (Internet Mail Access Protocol).

The basic difference between SMTP and POP3 is that **SMTP** is a *message transfer agent* used to send the message and **POP3** is a *message access agent* used to receive the message.

The SMTP is a protocol that defines the working of Message Transfer Agent. It is used to send the mail from the sender's computer to the receiver's mail server. POP is a pull protocol that is required to pull mail from receiver's mail server to receiver's computer. It also let user organize the mails.

E-mail Architecture



At the sender side

SMTP protocol is used in the scenario where both the **sender** and **receiver** of mail must be connected to their mail server by **WAN or LAN**. SMTP protocol is used two times, **first** between **sender and its mail server** and **second** between **client's mail server and receivers mail server**. SMTP is not used between receiver's mail server and receiver; the POP protocol accomplishes this task.

At the receiver side

To access mail from the mail box present at the mail server the **client MAA** at recipient computer establishes the **connection** with the mail server using **TCP** port **110**. For establishing connection *client* MAA at recipient's computer sends **username and password** to the mailbox. Then the user is authenticated to retrieve mail messages one by one.

Sender Side: The sender uses a **Mail User Agent (MUA)** (e.g., Outlook, Thunderbird) to create and send an email.

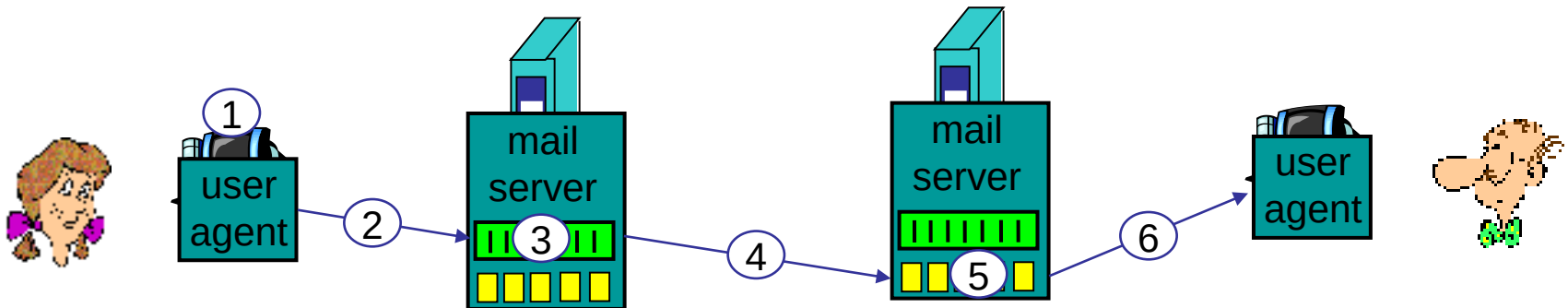
- The **Client MTA (Mail Transfer Agent)** on the sender's side sends the email to the sender's **Mail Server** using the **SMTP (Simple Mail Transfer Protocol)**.
- The **Server MTA** on the sender's mail server then forwards the email over the **Internet** to the receiver's mail server, also using **SMTP**.

Receiver Side: The receiver's mail server receives the email using **Server MTA** and stores it.

- The receiver's email client (MUA) retrieves the email from the mail server using **POP3 (Post Office Protocol 3)** or an alternative like **IMAP** (not shown here)
- The **Client MAA (Mail Access Agent)** handles the email download from the server to the receiver's machine.

Scenario: Alice sends message to Bob

- 1) Alice uses UA to compose message and “to” bob@some school.edu
- 2) Alice’s UA sends message to her mail server; message placed in message queue
- 3) Client side of SMTP opens TCP connection with Bob’s mail server
- 4) SMTP client sends Alice’s message over the TCP connection
- 5) Bob’s mail server places the message in Bob’s mailbox
- 6) Bob invokes his user agent to read message



SMTP Vs POP3

BASIS FOR COMPARISON	SMTP	POP3
Basic	It is message transfer agent.	It is message access agent.
Full form	Simple Mail Transfer Protocol.	Post Office Protocol version 3.
Implied	Between sender and sender mail server and between sender mail server and receiver mail server.	Between receiver and receiver mail server.
work	It transfers the mail from senders computer to the mail box present on receiver's mail server.	It allows to retrieve and organize mails from mailbox on receiver mail server to receiver's computer.

DNS

What is DNS?

DNS stands for **Domain Name System (DNS)** services. When we access a website, we are using this service to locate the server where the domain's website is located. When browsing the web, we usually type in a domain name like `www.google.com` into our browser. This is better than trying to remember an IP address linked to a Google server.

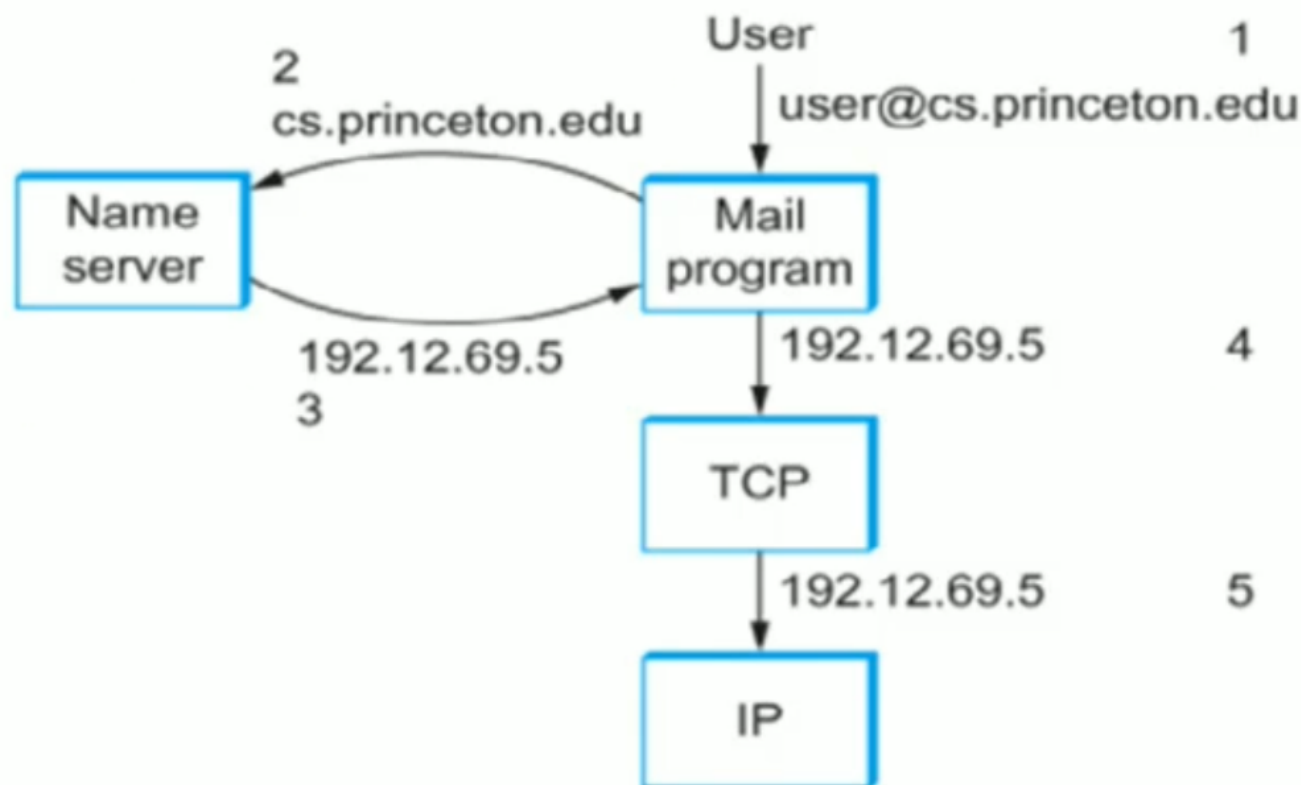
Behind the scenes, a conversion happens using this service which converts `www.google.com` to `172.217.12.46`. The IP address designates the location of a server on the Internet. This conversion process is called a query. This is an integral part of how devices connect with each other to communicate over the internet. To understand the query process, let's review how this query works.

The Domain Name System (DNS)

- Each Internet host is assigned a host name and an IP address
 - ▣ Host names are structured character strings, e.g., www.google.com
 - ▣ IP addresses are 32-bit integers
 - ▣ e.g., 139.130.4.5
- DNS is the naming service of the Internet that resolve host names to IP addresses
- DNS allows users of Internet applications to refer to remote hosts by name rather than by address



Host name translated to IP address



Host name translated into IP address. The numbers 1–5 show the sequence of steps in the process

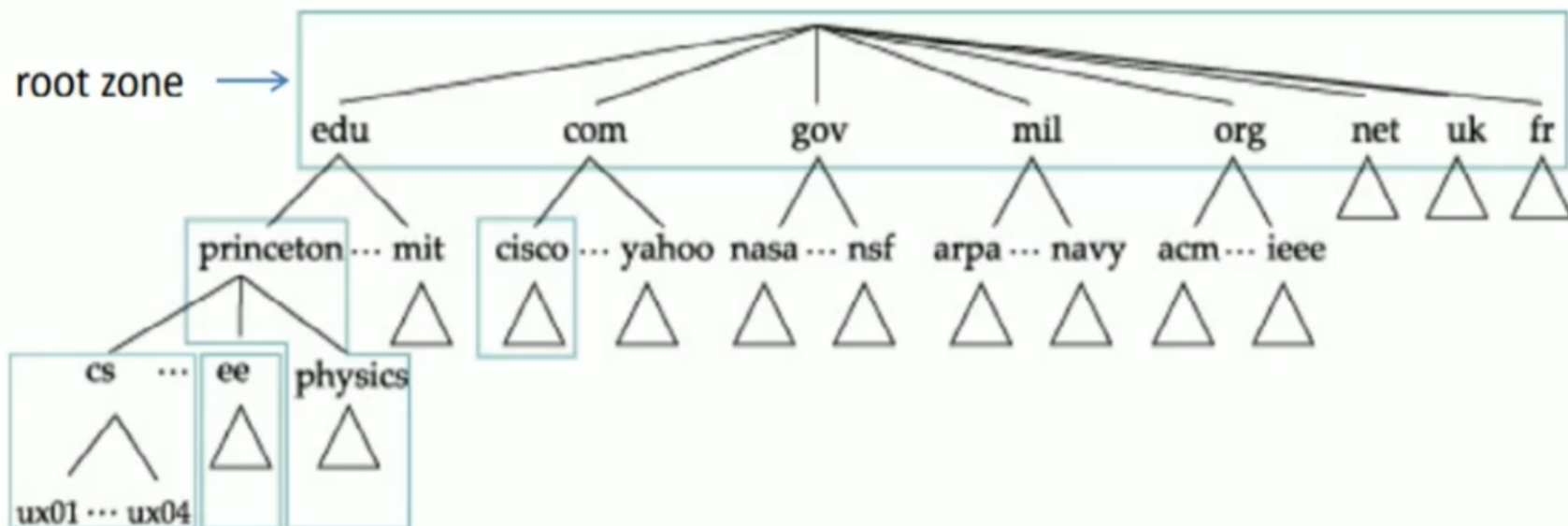
DNS Name Space

- Names in the DNS name space are called **domain names**
 - A domain name identifies a realm of administrative authority within the Internet
 - Domain names are **structured names**, i.e., they consist of multiple components
 - Example domain names: cs.iastate.edu, ieee.org, ox.ac.uk
- The DNS name space is represented by a tree with two types of nodes
 - Leaf nodes represent hosts
 - Non-leaf nodes represent domains
- Two types of top-level domains
 - Generic top-level domains: com, edu, gov, etc.
 - Country-code top-level domains: us, uk, fr, etc.

Generic Domains

Label	Description
com	Commercial Organizations
edu	Educational institutions
gov	Government institutions
org	Nonprofit Organizations
name	Personal names
aero	Airlines and aerospace companies





The DNS name space partitioned into zones

Resource Records

- Zone data are stored in resource records
- Each resource record is a 5-tuple (Name, Value, Type, Class, TTL)

Name Resolution

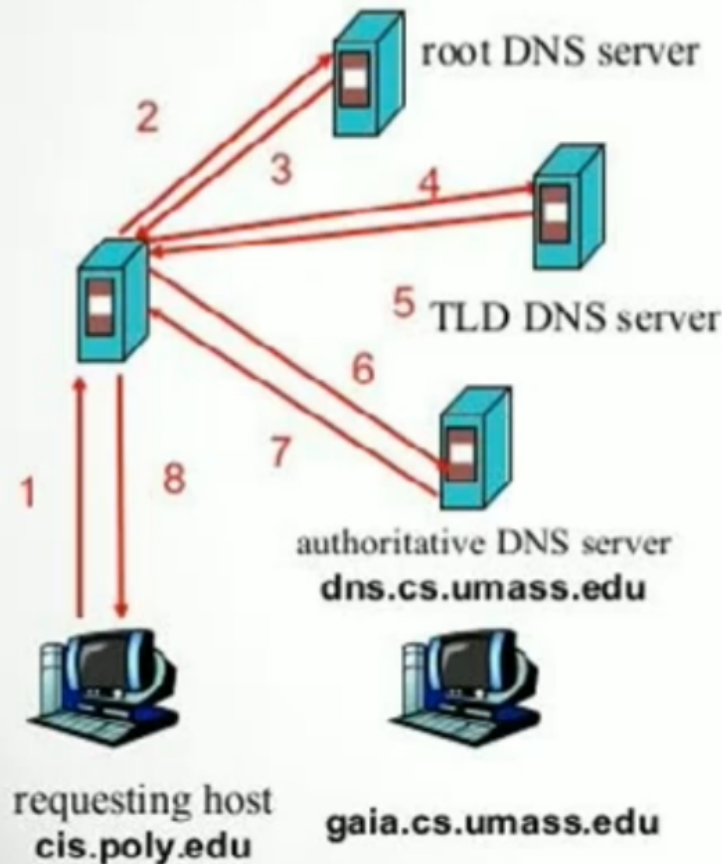
- The client program running on each Internet host is initialized with the address of a local name server
 - The local name server knows the address of one or more root name servers
- To resolve a name, a client sends a query to the local name server
 - The local name server queries other name servers on the client's behalf if it cannot resolve the name

Iterative vs. Recursive Name Resolution

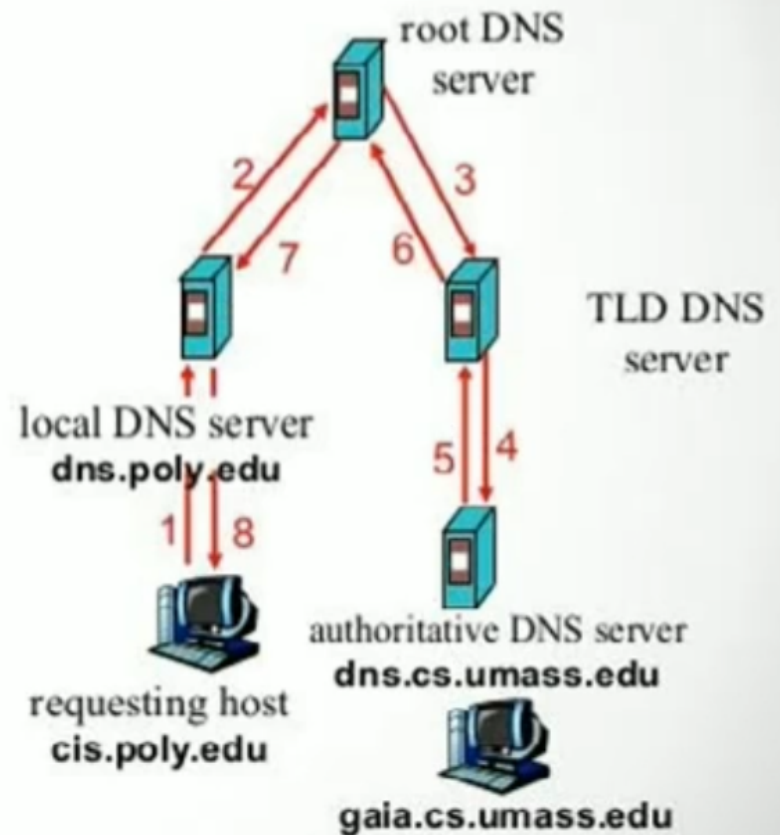
- In iterative name resolution, if the queried name server does not have a match for the queried name, it returns the IP address of a lower level name server
 - ▣ The local name server then queries the lower level name server; It continues this process until it locates a name server that can resolve the queried name
- In recursive name resolution, if the queried name server does not have a match for the queried name, it requests a lower level name server to resolve the queried name
 - ▣ The queried name server does not refer the local name server to a different name server as in iterative name resolution

DNS name resolution example

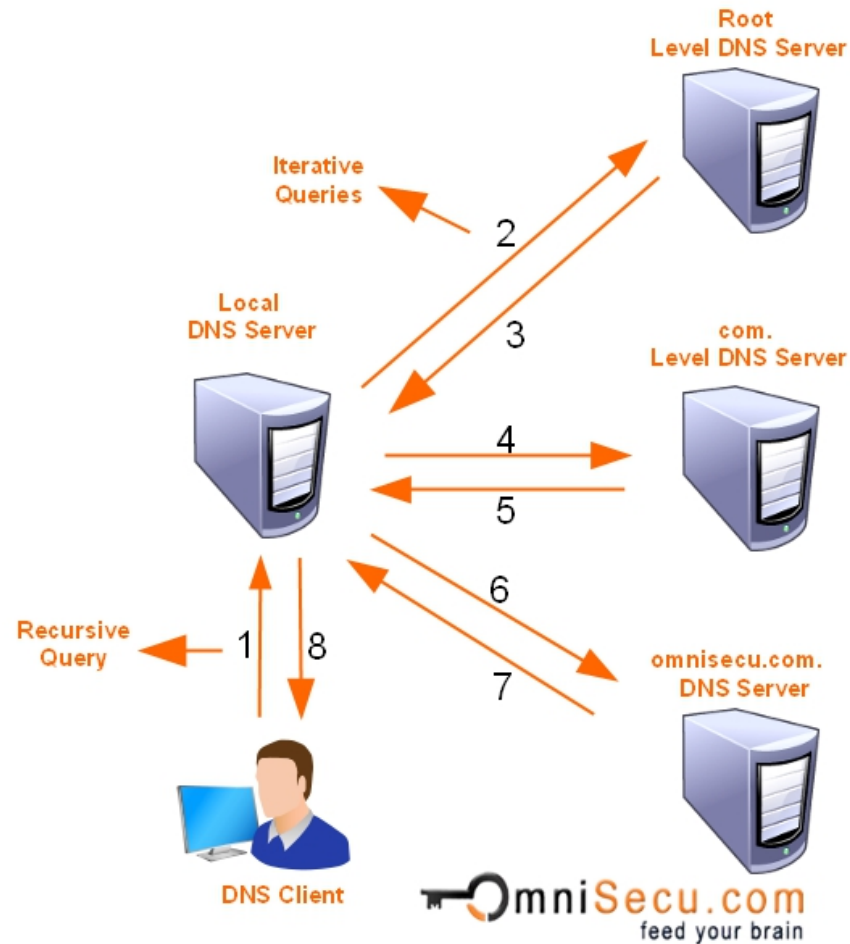
iterative query



recursive query



Iterative DNS Lookup



Recursive DNS Lookup

